

WiFi Backscatter for Audio Transmission with In-band Analog Modulation

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Abstract—In IoT applications such as environmental monitoring and industrial security surveillance, audio sensors are increasingly used, among which wireless sensors are preferred. To achieve a sustained transmission, low-power wireless technology such as backscatter has been widely considered. However, existing backscatter systems encounter difficulties in audio transmissions due to the high power consumption from the complicated digital processing and fast frequency-shifting clocks. In this paper, we propose Satori, the first-of-its-kind in-band analog backscatter system for audio transmission with ultra-low power consumption. Satori eliminates the need for in-place digital processing by directly embedding analog audio voltages into backscattered WiFi symbols through analog modulation. It also avoids the power consumption of the frequency-shifting clock by transmitting the audio within the excitation WiFi signal's band. We implement the Satori prototype and evaluate it under various settings. The results indicate that Satori can transmit audio at a sampling rate of 41.67 kHz and achieve a signal-to-noise ratio (SNR) exceeding 18 dB.

Index Terms—Backscatter, Analog, WiFi, Audio, RF Computing

I. INTRODUCTION

Audio sensors are increasingly used in IoT applications, such as environmental monitoring, smart cities, and factory security surveillance [1]–[3]. Since audio signals are susceptible to interference and attenuation, these sensors are typically positioned near sound sources and rely on wired or wireless transmission methods. In industrial settings, where cabling is costly and impractical, wireless sensors are preferred [4]. Generally, these wireless sensors are expected to transmit high-quality audio while minimizing power consumption.

Backscatter, as a low-power transmission technology [5]–[10], has shown potential in achieving audio transmission with minimal energy consumption. Significant efforts have been made toward developing battery-free wireless sensors [11]–[13]. However, the existing backscatter systems fall short in realizing the vision of ultra-low-power audio transmission due to the high energy requirements from the following two parts:

- **High-speed digital processing:** Existing backscatter systems typically rely on the analog-to-digital converter (ADC) and processors to digitize and process the audio signal. Constrained by this traditional digital-centric paradigm, these digital devices must operate at a high speed for high-quality

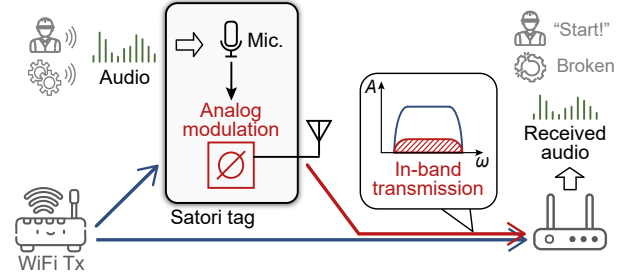


Fig. 1: Satori utilizes analog modulation and in-band transmission to transmit audio signals.

audio transmission tasks, resulting in an excessive power consumption burden on the backscatter tag.

- **Fast frequency-shifting clock:** Since the interference from the in-band excitation will seriously degrade the audio transmission quality, existing approaches often shift the backscattered signal to a different frequency band [6], [14]. However, the frequency shift necessitates a fast clock, leading to additional power consumption.

In this paper, we introduce Satori, the first-of-its-kind in-band analog backscatter system for audio transmission. Satori embraces the emerging paradigm of RF computing [15]. Unlike conventional digital-centric approaches, RF computing is a novel computational paradigm that treats RF signals as computational operands and processes information through direct manipulation of RF signals within the RF domain, thereby bypassing power-hungry digital conversions and unlocking possibilities for ultra-low-power sensing and communication.

Guided by this paradigm, the Satori tag is excited by the WiFi signal commonly available in industrial environments. As shown in Fig. 1, instead of digitizing the audio, it directly performs analog modulation in the RF domain. Specifically, it executes a phase mapping to embed the continuous analog audio signals into the phase of each WiFi symbol, thereby completely avoiding high-speed digital processing. Satori then transmits the audio directly inside the frequency band of the excitation WiFi signal, eliminating the fast and power-consuming frequency-shifting clock. By leveraging analog modulation and in-band transmission, Satori enables audio transmission with ultra-low power consumption.

However, realizing the idea is non-trivial and presents several technical problems. First, the backscattered signal is often overwhelmed by the strong in-band excitation and is prone to noise. Second, even a minor carrier frequency

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offset (CFO) induces phase variations in the received signal, which can completely obscure the backscattered signal. Third, the frequency-selective fading in the Tx-Tag-Rx channel can severely weaken the backscattered signal, degrading the audio transmission quality. Lastly, bursty WiFi transmissions cause the tag's modulation to fail when the excitation is absent.

Our prior work [16] establishes three core designs for Satori. At the tag end, we design in-band analog modulation, which directly embeds the analog audio voltage into WiFi symbols by shifting the reflected phase in the RF domain. We further optimize the modulation parameters to improve robustness against noise. At the receiver, we perform per-symbol channel estimation to extract the tag's modulation. We propose two-step CFO calibration: a coarse correction via phase rotation of the received signals, followed by symbol-by-symbol tracking based on cyclic prefix (CP). We also propose fading-adapted audio extraction, which quantifies the frequency-selective fading from the received signals and applies a weighted average across pilot subcarriers to mitigate the fading impact. To mitigate the data loss caused by bursty WiFi traffic, we propose a NUCZT-based audio spectrogram recovery method, which directly computes uniform time-frequency representations from the irregularly spaced audio frames, enabling reliable audio processing even under intermittent WiFi excitation.

We summarize our contributions as follows:

- 1) We present the in-band analog backscatter system for the first time. Guided by the RF computing paradigm, it achieves audio transmission with ultra-low power consumption, pushing the development of audio sensing in industrial environments.
- 2) We propose the design of in-band analog modulation, two-step CFO calibration, fading-adapted audio extraction and audio spectrogram recovery, addressing key technical challenges in the analog audio signal embedding and in-band transmission of backscattered signals.
- 3) We implement the Satori tag design on a printed circuit board (PCB) and the Satori receiver on a software-defined radio (SDR), and conduct evaluations in an indoor environment. The results indicate that Satori can transmit audio with an SNR exceeding 18 dB.

The rest of this paper is structured as follows: §II and §III present the tag and receiver designs. We propose the NUCZT-based spectrogram recovery in §IV and discuss practical deployment issues in §V. §VI describes the prototype, and we evaluate its performance in §VII. Finally, we review related work in §VIII and conclude this paper in §IX.

II. EMBEDDING THE AUDIO SIGNAL

This section introduces how the Satori tag achieves the in-band analog modulation, which physically realizes the RF computing paradigm. We first explain how the Satori tag continuously shifts the reflected phase according to the analog audio signal. To cancel out the strong interference, we propose the in-band analog modulation scheme and optimize its parameters for improving the robustness against noise. Finally, we present the design details of Satori tag's modulation circuit.

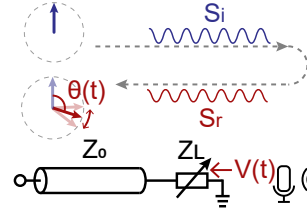


Fig. 2: The reflected phase shifts with the voltage.

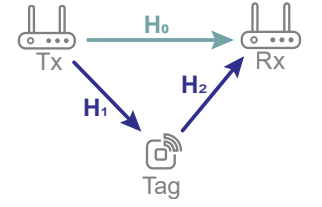


Fig. 3: The channel model of in-band backscatter.

A. Shifting the Reflected Phase

The Satori tag first uses an ultra-low-power microphone to convert the external audio signal to a voltage signal on the tag. This voltage signal is an analog signal that changes in real time with the external audio. We denote this analog voltage signal as $V(t)$, which carries the information of the audio. Following the RF computing paradigm, we consider using this analog voltage signal to control the tag's reflected phase, thereby injecting the audio information directly into the RF domain.

In an RF circuit, the reflected phase refers to the phase difference between the reflected signal S_r and the incident signal S_i , which is primarily determined by the terminal load impedance Z_L as:

$$\theta = \arg\left(\frac{S_r}{S_i}\right) = \arg\left(\frac{Z_L - Z_0}{Z_L + Z_0}\right), \quad (1)$$

where Z_0 is the characteristic impedance of the transmission line. By employing components with continuously tunable reactance as the terminal load, the reflected phase can be adjusted continuously. We directly feed the analog voltage $V(t)$ into the load, changing the load's reactance, and the reflected phase $\theta(V(t))$ is shifted by the voltage, as shown in Fig. 2. Thus, the tag can shift its reflected phase according to the analog audio signal in real time.

Generally, the reflected phase is expected to be linear with the voltage. This avoids voltage-dependent distortion and simplifies demodulation [11], [17].

B. In-band Analog Modulation Scheme

Modeling the Channel. The tag's modulation essentially alters the wireless channel between transmitter (Tx) and receiver (Rx). Considering the in-band backscatter scenario shown in Fig. 3, a tag shifts the reflected phase linearly with the voltage, tuning the Tx-Tag-Rx channel as follows:

$$H = H_0 + H_1 e^{j\theta(V(t))} H_2, \quad (2)$$

where H_0 , H_1 , and H_2 are the in-band excitation channel, the channel from WiFi Tx to the tag, and the channel from the tag to the Rx, respectively. $\theta(V(t))$ represents the tag's reflected phase varying linearly with the voltage $V(t)$. We represent the channel H as a vector in the in-phase/quadrature (IQ) plane, as shown in Fig. 4(a). When the voltage varies, the tag shifts its reflected phase, and the endpoint of the channel vector traces a circle, representing the tag's channel-tuning effect, which we call the "tuning circle". However, due to H_0 , the center of

this circle is displaced from the origin and unknown. To extract the tag's modulation effect from the channel changing, the tag should provide at least **three** points with distinct phases.

Designing the scheme. Since the interference and channel remain nearly static over short intervals, we encode information in relative phase differences across symbols to cancel self-interference. Thus, we design a scheme that encodes information into the relative phase changes. Specifically, the tag takes three backscattered WiFi symbols to carry an analog voltage. The first backscattered WiFi symbol is a **reference symbol**, with a fixed reflected phase, acting as a phase reference. The other two backscattered WiFi symbols have reflected phases that correspond to the current voltage value, which we refer to as **phase-shifted symbols**. Due to the wireless channels H_1 and H_2 introducing the same but unknown phase rotations to all three backscattered WiFi symbols, it is infeasible to represent the analog voltage using the phase of a single symbol. Instead, we focus on the phase differences between the phase-shifted symbols and the reference symbol, as shown below:

$$\Delta\theta_1(V(t)) = pV(t) + a_1, \quad \Delta\theta_2(V(t)) = pV(t) + a_2, \quad (3)$$

where pV represents the part of the phase difference that changes linearly with the analog voltage V , and p is the slope. a_1 and a_2 are additional fixed phase differences applied by the tag, preventing the three symbols from overlapping at some specific voltages.

Fig. 4(b) illustrates the phase diagram of the three symbols. Furthermore, we find that the differences of these three symbols form a triangle. We refer to it as a **tuning triangle**. The two internal angles $\varphi_1(t)$ and $\varphi_2(t)$ of the tuning triangle have the following relationship with $\Delta\theta_1(V(t))$ and $\Delta\theta_2(V(t))$:

$$\varphi_1(t) = \frac{1}{2}\Delta\theta_1(V(t)), \quad \varphi_2(t) = \frac{1}{2}(2\pi - \Delta\theta_2(V(t))), \quad (4)$$

Thus, the analog voltage value is carried in the tuning triangle formed by the three symbols. The tag periodically modulates the three WiFi symbols, continuously embedding the varying analog audio voltage. Since the tuning triangle is formed by the differences among three symbols in the IQ plane, even when in-band transmission is affected by interference from the original symbol, the differences between them remain unaffected. Thus, the tuning triangle robustly carries the voltage in in-band backscatter.

Optimizing the robustness against noise. To enhance noise robustness, we optimize the slope p and the fixed phase differences a_1 and a_2 in Eq. 3. A larger slope p increases phase variation and thus improves resilience to noise. To determine optimal a_1 and a_2 , we formulate an optimization problem: we uniformly sample N analog voltages $V[i]$ over $[0, V_0]$. Then, we simulate the modulation, add Gaussian noise to the three received symbols in the IQ domain and extract the analog voltage $\hat{V}[i]$. The goal is to minimize the extraction error between $\hat{V}[i]$ and $V[i]$.

$$\arg \min_{a_1, a_2} \sum_{i=1}^N (V[i] - \hat{V}[i])^2, \quad a_1, a_2 \in [0, 2\pi] \quad (5)$$

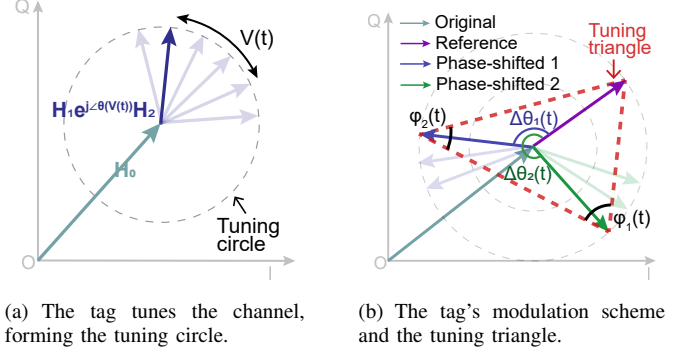


Fig. 4: The tag's channel tuning and modulation scheme, forming the tuning circle and the tuning triangle.

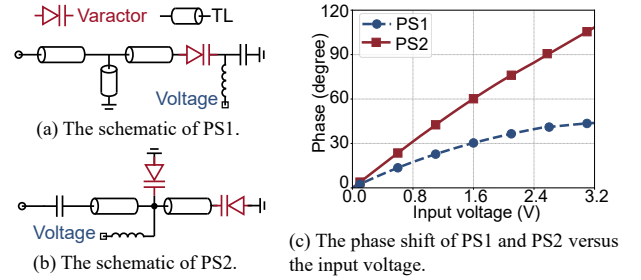


Fig. 5: Two kinds of reflective-type phase shifters and their performance comparison.

We first consider the theoretical derivation of the optimal parameters. However, it is challenging as illustrated in [16]. Therefore, we utilize a heuristic simulated annealing algorithm to solve the minimization problem. To avoid local optima, we run the algorithm multiple times with different initializations and select the best result. The optimized values are $a_1 = 0.5 \times (\pi - V_0/p)$ and $a_2 = 0.5 \times (3\pi - V_0/p)$.

C. Designing the Modulation Circuit

We first focus on how to modulate the reference symbols and phase-shifted symbols. We use a capacitor for the reference symbol. To modulate two phase-shifted symbols whose phases change linearly with the voltage, existing work proposes a reflective phase shifter (PS1) that consists of a varactor and three transmission lines (TL). When the diode is biased with the analog voltage, its capacitance changes, altering the reflected phase. Although this design extends the phase range via its three TLs, it renders the phase-voltage relationship nonlinear. [17] replaces one TL in Fig. 5 with another varactor diode. We further improve the design and propose PS2. In PS2, we compensate the nonlinearity via a second varactor together with optimized TL impedance and length. Using PathWave ADS, we optimize PS2 for phase range and linearity to meet the assumptions of Eq. 3. We limit the maximum phase shift to π to avoid phase ambiguity between the two phase-modulated symbols. We measure the phase within 20 MHz bandwidth using a vector network analyzer (VNA), and the maximum deviation is less than 5%.

However, high-frequency audio causes the microphone output voltage to change rapidly. For example, a voltage signal

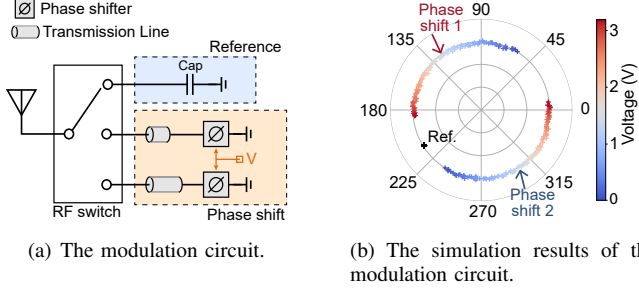


Fig. 6: The schematic and the simulation results of our reflective-type phase shifter.

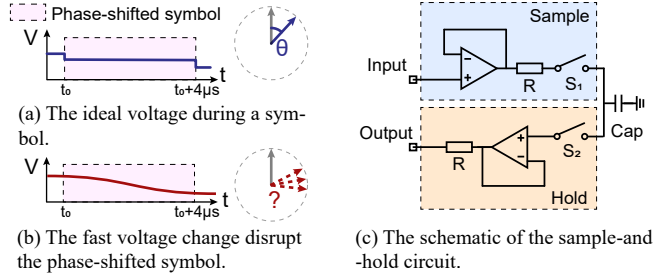


Fig. 7: The sample-and-hold circuit avoids the fast voltage changes disrupting the phase-shifted symbols.

with 4 kHz varies 10% of its amplitude in a WiFi symbol duration. Directly feeding the voltage signal into the phase shifter causes rapid phase variations within a symbol and corrupts the symbol, as illustrated in Fig. 7. Moreover, voltage drift between consecutive phase-shifted symbols distorts the tuning triangle and degrades voltage extraction accuracy. To address this, we introduce a sample-and-hold circuit (Fig. 7) acting as an analog memory module in the RF computing. It comprises two followers, two switches, and a capacitor. During the reference symbol, switch S_1 is closed and S_2 is open, allowing the capacitor to charge via the first follower. Before the phase-shifted symbols, S_1 opens and S_2 closes, enabling the second follower to maintain a stable output voltage based on the stored value.

Quick demonstration. Fig. 6(a) shows our modulation circuit. It consists of an RF switch and three loads. The first load generates the reference, and the other two are phase shifters. The analog voltage is stabilized by the sample-and-hold circuit before being fed to the phase shifters. We adjust the delay line length in front of the two phase shifters to ensure that the phase difference satisfies the optimized parameters. We use ADS to verify our design. Fig. 6(b) shows the reflection coefficients of the three loads under different voltage inputs, indicating that our design realizes the optimized parameters.

Finally, the low-power tag has a clock offset of hundreds of ppm, causing symbol synchronization error. Thus, we design the modulation mechanism of Satori tag, as shown in Fig. 8. When an excitation WiFi packet arrives, the tag skips the preamble and switches to the reference state in the middle of the first payload symbol to start modulating the reference symbol. Then, the circuit toggles the switch and modulates

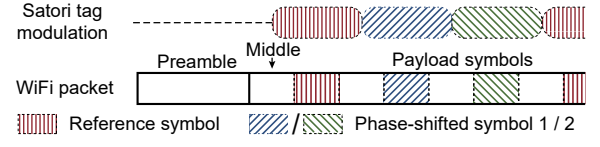


Fig. 8: Satori tag's modulation mechanism

the two phase-shifted symbols. The switch period is twice the WiFi symbol duration, ensuring that every even-index payload symbol is fully modulated. This mechanism tolerates up to 2 μ s of synchronization error (half a WiFi symbol), alleviating both symbol misalignment and tag frequency offset. Although this mechanism takes the time of six WiFi symbols to carry one voltage, it still achieves a sampling rate of 41.67 kHz with the WiFi symbol rate of 250 kHz, sufficient to support audio signal transmission.

III. EXTRACTING THE AUDIO SIGNAL

As shown in Fig. 9, the tag applies an identical phase shift to all subcarriers of a backscattered WiFi symbol. Following this operation, every three consecutive WiFi symbols construct a tuning triangle that encodes the analog voltage. Since the tag modulates the reflection phase on a per-symbol basis to embed audio, the Rx needs to recover baseband subcarriers of all symbols and perform per-symbol channel estimation to extract the modulation. We focus on pilot subcarriers, as their values are fixed and known across WiFi symbols.

The ideal extraction procedure at the Rx is as follows: upon receiving a packet, the Rx first conducts standard WiFi demodulation, then selects even-indexed payload symbols—those fully modulated by the tag—and extracts their pilot subcarriers. It then computes the inner angles of the tuning triangle from the IQ coordinates of pilot subcarriers across three adjacent symbols, and derives the analog voltage via Eq. 4 to recover the audio.

A. Two-step CFO Calibration

Although the receiving seems straightforward, the *carrier frequency offset* (CFO) and the *sample frequency offset* (SFO) can affect the phase. Among the two offsets, the CFO has a greater impact on the phase of the pilots [18]. Since the backscattered signal is only a small part in the received signal, CFO-induced phase rotation overwhelms the triangle formed by the pilots of the received symbols, as shown in Fig. 10.

The Rx uses the preamble for CFO calibration; there is still residual CFO (rCFO) in the payload [19]. Although the rCFO calibration has been studied in existing works, they can only be applied when the pilots are not modulated [19]–[21].

Considering that the rCFO fluctuates slightly around an average value, we propose the two-step CFO calibration method. It first estimates and removes the average rCFO, then tracks the residual fluctuation.

First step: average CFO correction For the first step, we focus on the reference symbols to detect and correct the average rCFO. The pilot subcarriers of these WiFi symbols can be modeled as:

$$R_k = (H_0 X + H_1 X e^{j\theta_{ref}} H_2) e^{2\pi j \cdot \text{rCFO} \cdot N_p k \tau}, \quad (6)$$

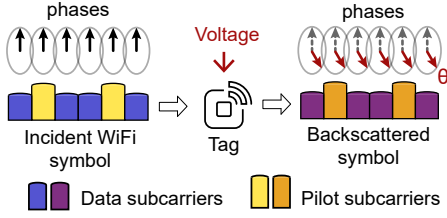


Fig. 9: The tag shifts the phases of all subcarriers of the backscattered symbol.

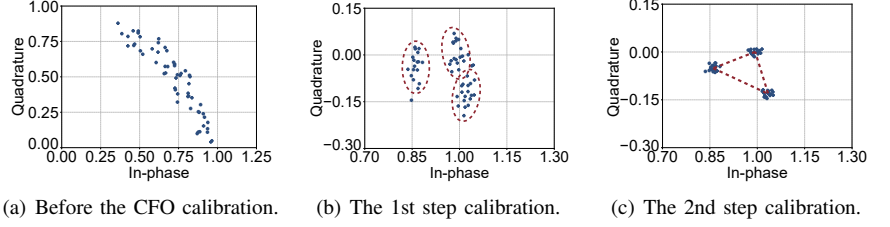


Fig. 10: The IQ points of the pilot subcarriers before and after the two-step CFO calibration.

where R_k is the pilot subcarriers of the k -th reference symbol. H_0 , H_1 , and H_2 represent the wireless channel as described in Fig. 4(a). X , θ_{ref} , and τ denote the pilot in the original WiFi symbol, the tag's phase change, and the duration of the WiFi symbol, respectively. N_p is equal to 6, as the interval between every two reference symbols is 6 WiFi symbols.

Since X and θ_{ref} are fixed, and H_0 , H_1 , and H_2 are stable within a WiFi packet, there is a linear relationship between the phase of R_k and rCFO as follows:

$$\angle R_k = \angle(H_0 X + H_1 X e^{j\theta_{ref}} H_2) + 2\pi \cdot \text{rCFO} \cdot N_p k \tau, \quad (7)$$

where $\angle(H_0 X + H_1 X e^{j\theta_{ref}} H_2)$ is a constant phase.

We use the linear fitting method to estimate the average rCFO and complete the first step of CFO calibration. As shown in Fig. 10(b), after the first step, the received pilot subcarriers converge into clusters but still fail to form a clear tuning triangle for voltage extraction. This is because the fluctuating part of the rCFO cannot be finely calibrated in the first step.

Second step: fluctuating CFO calibration The second CFO calibration step uses the CP section within each WiFi symbol to track and calibrate the fluctuating part of the rCFO. For WiFi symbols sampled at 20 MHz, the CP consists of the first 16 samples, which are the same as the last 16 samples. This allows us to detect the rCFO level as follows:

$$\hat{\text{rCFO}}_n = \frac{N_s}{F_s} \angle \left(\sum_{t=1}^{16} (x_n^*[t] x_n[t + N_s]) \right), \quad (8)$$

where $\hat{\text{rCFO}}_n$ and $x_n[t]$ are the rCFO and the t th sample in the n th WiFi symbol. And N_s represents the number of samples in the interval between the CP and the last 16 samples, which is 64 in WiFi orthogonal frequency-division multiplexing (OFDM). F_s is the sampling rate, which is 2×10^7 . We use a smooth update method to estimate the $\hat{\text{rCFO}}_n$ and apply it to calibrate the n th WiFi symbol, yielding the results shown in Fig. 10(c). Thus, the two-step calibration effectively removes the rCFO, yielding a clear tuning triangle.

One may be concerned that the SFO between the Tx and the Rx could also impact audio extraction. For a certain pilot, the SFO-induced phase rotation is also linearly related to the symbol index [19] and can be calibrated in the first step.

B. Fading-adaptive Audio Extraction

After the calibration, the Rx can extract the tuning triangle from each pilot subcarrier to extract the voltage. In real

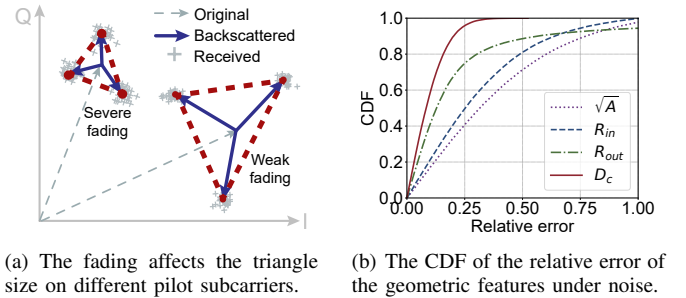


Fig. 11: Use the geometric features to quantify the fading.

environments, frequency-selective fading in the Tx-Tag-Rx channel can severely attenuate certain pilot subcarriers of the backscattered signal, leading to tuning triangles that are highly susceptible to noise. Given that a 20 MHz WiFi signal includes four pilot subcarriers, we can extract the analog voltage from each triangle and fuse them via weighted averaging:

$$\hat{V}(t) = \sum_{p=1}^4 w_p \hat{V}_p(t) / \sum_{p=1}^4 w_p, \quad (9)$$

where $\hat{V}_p(t)$ is the analog voltage estimated from the p th pilot subcarrier based on Eq. 4 and Eq. 3, and w_p denotes the weight assigned to that subcarrier. Subcarriers with stronger fading receive smaller weights, reducing their contribution.

However, in in-band transmission, the backscattered signal overlaps with the direct WiFi signal, obscuring the fading estimate. To address this, we propose a fading-adaptive audio extraction method. We observe that deep fading reduces the amplitude of affected pilots, shrinking the tuning triangle shown in Fig. 11(a).

We consider using the radius of the triangle's circumcircle R_{out} , as the weight w_p , as it is equal to the amplitude of the backscattered pilot. However, the R_{out} is highly sensitive to noise and may deviate considerably from the actual pilot amplitude under noisy conditions. To improve robustness, we explore alternative geometric features of the tuning triangle, such as the square root of the triangle area $\sqrt{A}$, the inradius R_{in} , and the total distance from the centroid to the three vertices D_c . All these features scale with the backscattered symbol amplitude and can serve as weights w_p .

To assess their noise sensitivity, we conduct a Monte Carlo analysis: we construct an ideal triangle, add Gaussian noise to its vertices, and compute the relative error of each geometric

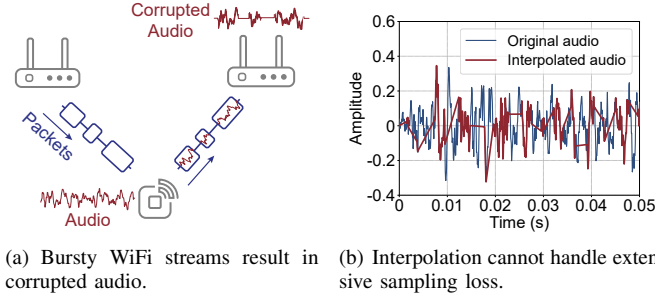


Fig. 12: Interpolation methods cannot adapt to the sampling loss caused by bursty WiFi streams.

feature with respect to the ideal case. In Fig. 11(b), we show the CDF of the relative error for each geometric feature after adding the Gaussian noise to the tuning triangle. We observe D_c has both a lower median and 95% relative error than other features. This shows D_c quantifies pilot fading more accurately under noise. We select D_c as w_p , and eventually extract the analog audio signal using Eq. 9.

IV. RECOVERING THE AUDIO SPECTROGRAM

Unlike RFID systems with a continuous carrier, WiFi transmission is bursty and has a varying duty cycle. When WiFi excitation is absent, Satori's audio sampling is lost, as shown in Fig. 12(a). When the WiFi duty cycle decreases, the proportion of lost audio samples increases, leading to a significant degradation in the quality of the extracted audio. Consequently, such an audio signal becomes unsuitable for downstream tasks.

Given that WiFi traffic can be regarded as a random Poisson stream, the audio signal transmitted by Satori effectively represents a randomly sampled version of the original audio signal. A straightforward method is to interpolate missing samples in the time domain. However, this method is only effective when the proportion of missing samples is relatively low. As shown in Fig. 12(b), when the WiFi duty cycle drops to 30%, simple spline interpolation results in a considerable error compared to the original audio, indicating that the lost temporal information cannot be accurately reconstructed through interpolation alone.

Since many audio sensing tasks rely on some spectral features (e.g., STFT, Mel spectrogram, and MFCC), we consider recovering the audio spectrogram. From another perspective, leveraging the prior knowledge that audio information is often more concentrated in the low-frequency domain allows us to more effectively utilize Satori's sampled audio to reconstruct the original audio's spectrogram as accurately as possible.

We define the problem as the spectrogram reconstruction of a band-limited signal from random samples. Some existing approaches consider compressed sensing (CS) techniques; however, compressed sensing requires the signal to have a sparse spectrogram. In contrast, general audio signals, while band-limited, are not inherently sparse in the frequency domain. To address this issue, we first consider a Non-Uniform Discrete Fourier Transform (NUDFT), which enables spectral

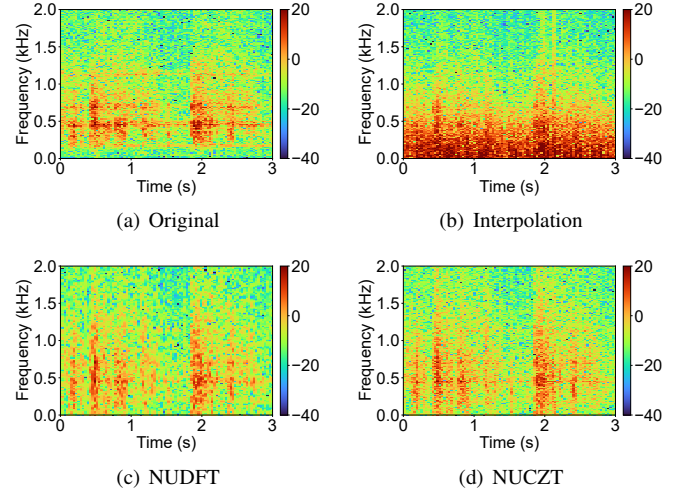


Fig. 13: The original spectrogram and the spectra recovered by interpolation, NUDFT, and NUCZT when duty cycle is 30%.

computation for non-uniformly sampled time-domain signals. The NUDFT can be expressed as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-2\pi j \frac{kt[n]}{N}}, \quad (10)$$

where $x[n]$ represents the n -th randomly sampled point of the Satori audio signal, $t[n]$ denotes the corresponding sampling time, and N is the number of samples within a time window. Similar to STFT, we segment the extracted audio signal using a sliding window. In each window, we perform NUDFT, then concatenate the results across all windows to reconstruct the spectrogram from the randomly sampled signal.

However, while NUDFT can recover the signal spectrogram, its spectral resolution Δf is defined as $\Delta f = \frac{f_s}{N}$, where f_s is the sample rate of Satori (i.e., 41.67 kHz). When the WiFi duty cycle is low, Satori transmits fewer sampling points within a time window, leading to a lower spectral resolution in NUDFT. As shown in Fig. 13(c), the NUDFT-reconstructed spectrogram appears blurred, failing to capture the fine details.

Noting that audio signals are primarily concentrated in the low-frequency range, we adopt the Chirp-Z Transform (CZT). CZT is expressed as $X[k] = \sum_{n=0}^{N-1} x[n] (AW^{-k})^n$, where A and W are constants. We define $A = 1$ and $W = e^{-2\pi j \frac{p}{N}}$, where p is a spectral scaling factor less than 1. According to the definition of CZT, $X[k]$ represents a uniform N -point sampling of the spectrogram within the frequency range $(-\frac{f_s p}{2}, \frac{f_s p}{2})$, with the improved spectral resolution of $\frac{f_s p}{N}$. This method prioritizes spectral resolution in the low-frequency range, where most audio information is concentrated, while allocating fewer resources to the high-frequency components, which typically carry less perceptually significant information. We further modify the CZT to accommodate non-uniformly sampled time-domain signals and propose the Non-Uniform Chirp-Z Transform (NUCZT):

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-2\pi j \frac{kpt[n]}{N}}, \quad (11)$$

As shown in Fig. 13(d), since the NUCZT method enhances the spectral resolution, it preserves more details of the original spectrogram, improving its utility for audio sensing tasks.

Regarding the selection of spectral scaling factor p in NUCZT, we first use NUDFT to obtain a coarse estimate of the spectral energy distribution. We then choose p such that 95% of the total energy falls within the range $(-\frac{f_s p}{2}, \frac{f_s p}{2})$, thereby adapting to different types of audio signals. Furthermore, the time-domain audio signal can be recovered through the Inverse Chirp-Z Transform [22].

V. PRACTICAL ISSUE

We complete the design by addressing some practical issues.

A. Phase Shifter: Linearity vs. Phase Range

In the design of PS2, we optimize the properties of TL in the ADS to achieve high linearity and a wide phase shift range. However, in practical design, although the mutual compensation of two varactors can partially offset nonlinearity, a larger phase shift range often leads to the deterioration of linearity, thereby violating the assumption of linear phase shift in Eq. 3. This presents a trade-off issue.

To address this problem, our optimization process does not simultaneously target both linearity and phase shift range. Instead, we first ensure that the Pearson correlation coefficient (r) between the voltage and phase shift in the phase shifter exceeds 0.99. Subsequently, we adjust the TL parameters to achieve a larger phase shift. Ultimately, our designed phase shifter guarantees a linearity of $r = 0.992$ and a phase shift of 108° within the voltage range of 0 to 3.3 V.

B. Sample-and-Hold Circuit: Follower Isolation

In the sample-and-hold circuit, we employ the followers and the analog switch to store the audio voltage during the reference symbol period in a capacitor. However, due to the limited driving capability of the low-power follower, directly connecting the follower to the capacitor causes jitter and prevents the voltage from stabilizing within one symbol period.

To address this issue, we introduced an isolation resistor between the follower and the capacitor. The function of this isolation resistor is to shift the pole of the entire circuit to a higher frequency, thereby providing sufficient phase margin to avoid jitter. Specifically, the value of the isolation resistor is determined as follows:

$$R_{iso} = \frac{1 + \sqrt{1 + (8\pi \cdot R_o \cdot C \cdot f_{GBW})}}{4\pi \cdot C \cdot f_{GBW}}, \quad (12)$$

where R_o and f_{GBW} are the output resistance and gain-bandwidth product of the follower, and C is the value of the capacitor. Ultimately, we select a 33 pF capacitor and a 470 Ω isolation resistor, enabling the capacitor voltage to stabilize in 4 μ s.

C. Envelope Detector: Off-Band Interference Suppression

We employ an envelope detector to detect the arrival of WiFi packets. However, a simple envelope detector converts all RF signal energy into voltage without frequency discrimination. Thus, signals from other bands could also trigger the envelope detector, erroneously activating the tag.

We place a surface acoustic wave (SAW) filter in front of the envelope detector. This filter introduces only about 2 dB of insertion loss for signals in the 2.4 GHz band while providing at least 30 dB of attenuation for signals in other frequency bands, significantly reducing the probability of false triggers. For other WiFi signals within the 2.4 GHz band, the CSMA/CA mechanism ensures that two nearby Tx do not operate simultaneously, so the tag's detection of Tx signals remains unaffected. Additionally, the group delay introduced by the filter is only 100 ns, which does not significantly increase the delay.

VI. IMPLEMENTATION

Tag: We implement the Satori tag's prototype on a 4-layer PCB using commercial off-the-shelf components, as shown in Fig. 14. This prototype consists of three modules: the packet detector, the modulation circuit, and the logic unit. Both the packet detector and the modulation circuit are connected to standard 2.4 GHz omnidirectional antennas with 6 dBi gain and are linked to the logic unit.

The packet detector is implemented using an envelope detector LT5534 and a comparator TLV3201. The Satori tag uses it to detect the WiFi packet and start modulation.

The modulation circuit comprises two components: a sample-and-hold module and a reflective module. An external low-power microphone provides the analog voltage input to the sample-and-hold module, which employs two ultra-low-power voltage followers TLV9041 and a CMOS switch TS5A3159 for toggling between sampling and hold modes. The stabilized output is then fed to the reflective module, which includes three load options: two reflective-type phase shifters constructed using microstrip lines and varactor diodes MAVR-000120, and a fixed 10 pF capacitor. Load switching is achieved using two RF switches SKY13335.

The control unit uses a low-power GW1N-LV4 FPGA. This unit's function only includes checking the output from the detector and controlling the modulation circuit without performing any digital computation or processing.

WiFi Transceiver: We implement the WiFi Tx using the USRP N210 software-defined radio platform. We generate the IEEE 802.11n compliant baseband signal with MATLAB's Wireless Waveform Generator App on a PC and send it to the USRP N210 through GNU Radio. The Tx includes an RF power amplifier that boosts the signal's average power to 20 dBm, complying with local regulations on civil radio. We use the USRP B210 platform for the Rx. We control the USRP B210 with PicoScenes [23] software on a laptop to receive the signal and perform audio reception in MATLAB. Both the Tx and Rx are equipped with a 2.4 GHz omnidirectional antenna with 6 dBi gain.

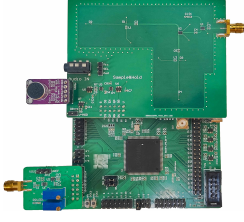


Fig. 14: The prototype of the Satori tag.

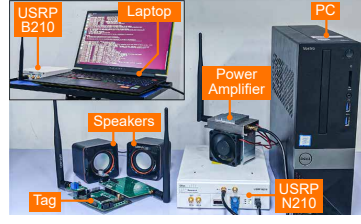


Fig. 15: Experiment setup of the evaluation.

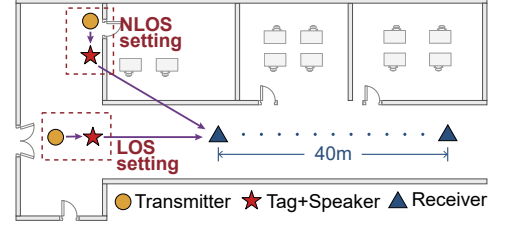


Fig. 16: Floorplan of the experiment environment for LOS and NLOS evaluation.

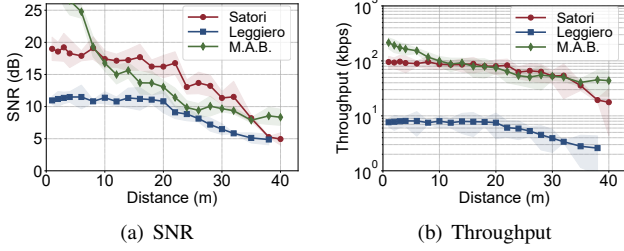


Fig. 17: SNR and throughput performance versus Tag-to-Rx distance in the LOS setting.

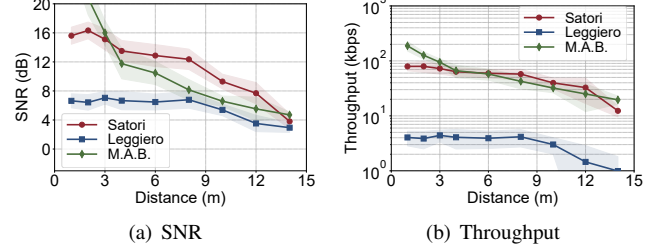


Fig. 18: SNR and throughput performance versus Tag-to-Rx distance in the NLOS setting.

VII. EVALUATION

A. Experiment Methodology

We deploy the WiFi Tx, Rx, and the Satori tag in a corridor, with two speakers playing audio signals near the tag. We evaluate Satori in both line-of-sight (LOS) and non-line-of-sight (NLOS) scenarios, as shown in Fig. 16. The Tx transmits WiFi packets at a default rate of 2000 packets per second, with a default duty cycle of 80%. The Tx's transmission power is set to 20 dBm. The speakers play the default audio that covers frequencies from 200 Hz to 4000 Hz.

To evaluate the audio transmission quality of Satori, we first record the output of the microphone on the tag at a bitrate of 320 kbps, which serves as the original audio signal. Since numerous audio analysis applications are based on the audio spectrogram, the primary metric in our evaluation is the **SNR** of the audio spectrogram recovered by the Rx, with respect to the original audio spectrogram. Meanwhile, we evaluate the **throughput** of Satori to quantify the audio transmission capacity. We sample the received analog audio into a bit sequence, and only the bits that are continuously correct from the most significant bit are considered valid. The throughput is then calculated as the number of valid bits received per second.

We take Leggiero [11] and the Microphone Array Backscatter (M.A.B.) [24] as baselines. The former is an analog backscatter system based on ambient WiFi, and the latter is the state-of-the-art analog backscatter for audio sensing. The Tx provides a single-tone carrier for M.A.B., and the tag modulates 1-track audio on the carrier.

We evaluate the SNR and throughput of Satori across different Tag-to-Rx distances under LOS and NLOS settings in §VII-B. We also evaluate the impact of audio frequency, audio strength, the duty cycle of the WiFi traffic, and tag synchronization error on SNR in §VII-C. Furthermore, we

conduct ablation studies by removing the sample-and-hold circuit, CFO calibration, fading-adaptive audio extraction in §VII-D. Finally, we demonstrate the proof-of-concept applications in §VII-G.

B. Overall Performance

We evaluate the SNR of the recovered audio spectrogram under varying Tag-to-Rx distances, with Tx-to-Tag distances set to 0.5 m, in both LOS and NLOS environments. Fig. 17 presents the SNR and throughput results in LOS scenarios, with shaded regions indicating variance.

First, Satori achieves SNRs above 18 dB at short Tag-to-Rx distances, corresponding to approximately 3 effective bits per sample. It achieves a throughput of 99.5 kbps. As distance increases, the SNR and throughput decrease. Nevertheless, Satori maintains an SNR of 5 dB and a throughput of 20.6 kbps at distances up to 40 m, sufficient for basic audio tasks.

Second, compared to Leggiero, Satori consistently achieves 5–10 dB higher SNRs when the Tag-to-Rx distance is within 30 m. The throughput of Satori is more than 10 times that of Leggiero. This is because Satori's symbol-level analog embedding allows for a higher audio sampling rate, while Leggiero's packet-level embedding limits its bandwidth.

Third, the SNR performance of Satori is comparable to or better than M.A.B. across most distances. M.A.B. outperforms Satori at short ranges because of Satori's residual self-interference, while M.A.B. avoids this issue via frequency shifting. At greater distances, Satori surpasses M.A.B. by over 5 dB, owing to its robust backscatter design. Given M.A.B.'s higher sampling rate of 50 kHz, its throughput also exceeds Satori within 10 m. At distances of 10–30 m, Satori achieves comparable throughput to M.A.B. due to its higher SNR.

Fig. 18 shows the SNR in NLOS scenarios. Leveraging fading-adaptive audio extraction, Satori achieves a maximum

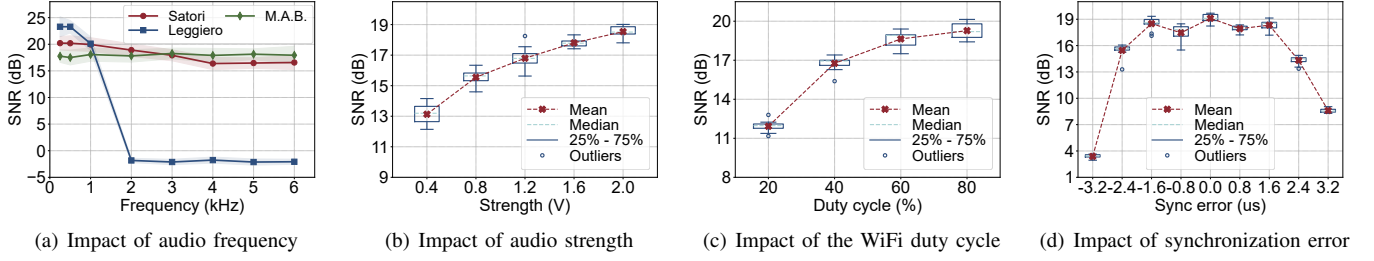


Fig. 19: Impact of audio frequency, audio strength, WiFi duty cycle, and synchronization error on the SNR.

SNR of 16 dB and a throughput of 82.9 kbps. It supports communication distances exceeding 12 m. We also tested Satori's performance at longer Tx-to-Tag distances, which can be found in our conference paper [16] presented in ACM MobiSys 2025.

C. Impacting Factors

We set the Tag-to-Tx distance to 0.5 m and the Tag-to-Rx distance to 8 m in the LOS scenario. We evaluate the impacting factors, including the audio frequency, the audio strength, the WiFi traffic duty cycle, and the synchronization error.

1) *Audio Frequency*: We play a narrowband audio signal with center frequency ranging from 250 Hz to 6000 Hz to the Satori tag and the two baselines, and evaluate the SNR of the recovered audio spectrogram.

As shown in Fig. 19(a), Satori exhibits decreasing SNR with increasing frequency, with the SNR at 6000 Hz being 4 dB lower than at 250 Hz. This is primarily caused by the performance degradation of the follower in the sample-and-hold circuit at high frequencies, while the NUCZT spectrum recovery algorithm is not sensitive to the frequency of the signal. In contrast, M.A.B. maintains a stable performance across all frequencies. Leggiero shows a sharp SNR drop beyond 1 kHz, limited by its packet-level embedding rate.

2) *Audio Strength*: We evaluate how audio strength affects the recovered spectrogram SNR, using the microphone's peak-to-peak voltage (Vpp) as the measure of signal strength. Results are shown in Fig. 19(b).

As audio strength increases, the SNR also improves, since higher amplitudes induce larger phase shifts in the backscattered signal, enhancing robustness to noise. Notably, when the audio strength drops from 2.0 Vpp to 0.4 Vpp and the audio signal energy drops by 14 dB, the SNR decreases by only 6 dB, demonstrating Satori's resilience to varying audio amplitudes.

3) *Duty Cycle of WiFi Traffic*: We evaluate the impact of different WiFi traffic duty cycles from 20% to 80% on the recovery of test audio. The results are shown in Fig. 19(c).

As the duty cycle decreases, Satori's SNR also decreases. At a duty cycle of 80%, Satori achieves approximately 19 dB SNR, while at 20%, the SNR drops to around 11.5 dB. Besides, NUCZT degrades more gracefully than interpolation and NUDFT at low duty cycles. This is because interpolation loses more samples as the duty cycle decreases, leading to a sharp drop in audio quality. While NUDFT can handle irregular sampling, its spectral resolution also degrades with lower duty cycles, resulting in reduced SNR.

4) *Synchronization Error*: We evaluate the impact of synchronization errors on the recovered spectrogram SNR, as Satori requires precise embedding at specific WiFi packet positions. In this evaluation, we vary the synchronization error from 0 to $\pm 3.2 \mu\text{s}$. The results are presented in Fig. 19(d).

When the synchronization error is within $1.6 \mu\text{s}$, the recovered spectrogram SNR is almost unaffected, remaining above 18 dB. When the error exceeds $2.4 \mu\text{s}$, the SNR drops rapidly. This is due to the tag's toggling cycle being twice the WiFi symbol duration, such that synchronization errors under half the WiFi symbol period allow for full modulation and audio recovery. Thus, Satori is robust to synchronization errors resulting from ultra-low-power packet detection.

D. Ablation Study

We conduct the ablation study by removing the sample-and-hold circuit, CFO calibration, and the weight in fading-adaptive audio extraction, assessing each component's impact on audio transmission.

1) *Effectiveness of the Sample-and-Hold Circuit*: We play narrowband audio with center frequency ranging from 250 Hz to 6000 Hz to Satori, evaluating the recovered spectrogram SNR with and without the sample-and-hold (S&H) circuit. The Tx-to-Tag and Tag-to-Rx distances are 0.5 m and 8 m. Results are shown in Fig. 20. When audio frequencies are below 2000 Hz, the SNR is similar with or without S&H. At higher frequencies, omitting S&H leads to a more than 2 dB SNR drop. The S&H stabilizes the voltage during each WiFi symbol, enhancing high-frequency audio quality.

2) *SNR Enhancement through the CFO Calibration*: We evaluate CFO calibration by comparing Satori under three modes: full calibration, rough calibration (only first step), and no calibration. With Tx-to-Tag distances of 0.5 m, we vary the Tag-to-Rx distance and compare the recovered audio SNR, as shown in Fig. 21. The full CFO calibration achieves the highest SNR, followed by rough calibration, with no calibration performing worst. At closer Tag-to-Rx distances, full calibration improves SNR by about 3 dB over no calibration. As the distance increases, the improvement from full calibration exceeds 7 dB, since CFO has a stronger impact on weaker backscattered signals, making calibration more critical for audio recovery.

3) *Improvement from the Weight in Audio Extraction*: We evaluate the impact of subcarrier weighting in fading-adaptive audio extraction by comparing the original weighted method

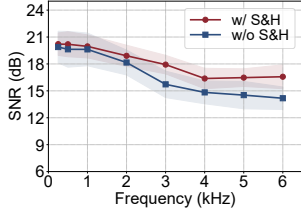


Fig. 20: The SNR enhancement from the sample-and-hold circuit.

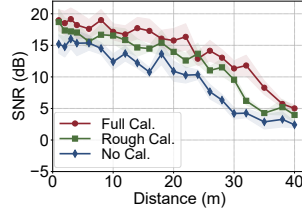


Fig. 21: The SNR enhancement from the CFO calibration.

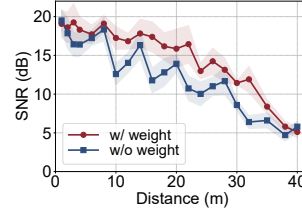


Fig. 22: SNR enhancement from the weight in the fading-adaptive audio extraction.

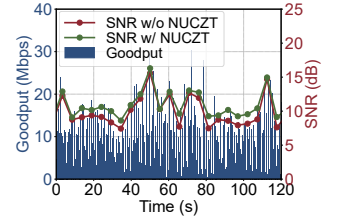


Fig. 23: The SNR performance of Satori in a real WiFi bursty traffic.

TABLE I: The power consumption of the PCB prototype and ASIC simulation.

Part	Logic unit	Packet detection	Clock
PCB	8.9 mW	23.9 mW	12.9 mW
ASIC	2 μ W	6.2 μ W	1.6 μ W
Part	S&H circuit	Switch	Total
PCB	33 μ W	31 μ W	45.6 mW
ASIC	2.3 μ W	2 μ W	14.1 μ W

(w/ weight) with a modified version using equal weights (w/o weight). The SNR of the recovered spectrogram is measured in both LOS and NLOS scenarios. As shown in Fig. 22, the w/ weight method consistently achieves higher SNR. At closer distances, the improvement is less than 3 dB. As distance increases, frequency-selective fading becomes more severe, and the w/ weight method better adapts to this fading, resulting in an SNR improvement of over 5 dB compared to w/o weight.

E. Operating in Real WiFi Traffic

To assess the performance of Satori in real WiFi networks with bursty traffic, we use the commercial network interface card (NIC) QCA9300 to transmit burst packets within 2 minutes. We replay the WiFi traffic generated during a real-time video streaming session on the NIC. We record the WiFi goodput 3 times per second and measure the SNR of the audio segments recovered by Satori with 5-second intervals. The measurement results are presented in Fig. 23. The SNR of the audio recovered by Satori varies in real-time with the WiFi throughput. This is mainly because a lower throughput often implies a lower duty cycle, which leads to a decrease in the quality of the audio recovered by Satori. We also notice that Satori with NUCZT can achieve higher audio SNR when the WiFi traffic is low.

F. Power Consumption

We evaluate the power consumption of the PCB prototype of the Satori tag and each of its components. The components we evaluated include the logic unit of the Satori tag, clock generation, packet detection, the sample-and-hold circuit, the RF switches, and the phase shifters. We measure the current using multimeter [25]. Since the error of the multimeter is 0.1 μ A, we use the datasheet to estimate the power if the component draws current less than 0.1 μ A.

The FPGA core consumes 8.9 mW, and the 1 MHz clock generation draws 3.9 mA, consuming 12.9 mW. The packet

detection module mainly relies on the LT5534, while the comparator TLV3201 adds only 120 μ W. In the sample-and-hold circuit, TLV9041 voltage followers consume 33 μ W due to alternating operation. The two analog switches draw less than 20 nA each [26], totaling 66 nW. Two RF switches consume 31 μ W. The reverse-biased varactors in the phase shifters draw only 1.6 pA [27] and can be ignored.

We further simulate the ASIC solution. We use Cadence Virtuoso for the analog part (e.g. S&H circuit) and Synopsys EDA for the digital part (e.g. logic unit). The design is synthesized using TSMC 65 nm technology standard cell library at the Typical-Typical (TT) process corner with a temperature of 25°C and a core voltage of 1.2 V. The post-synthesis power estimation of each component is shown in Tab. I. Benefiting from the 1 MHz low clock rate and passive voltage-to-phase conversion, the Satori tag avoids the hundreds of microwatts required for digital processing and the tens of MHz clock needed for frequency shifting, reducing the power consumption to approximately 14 μ W.

G. Proof-of-Concept Application

We demonstrate Satori's applications in command recognition. We use 12 common control commands in the Google speech commands dataset [28]. We input these audio commands into Satori, and classify via the ResNet34 network [29]. We first set the duty cycle to 70% and vary the Rx-to-Tag distance to measure recognition accuracy and the maximum and minimum accuracy rates among all commands. Then, we fixed the distance and varied the duty cycle, measuring the average accuracy. For the latter, we further compare it with interpolation and NUDFT spectrogram recovery methods.

Fig. 24 shows the accuracy rate with different Tag-to-Rx distance. The average accuracy can reach 95% when the Rx is 15 m away from the tag. When the Tag-to-Rx distance is shorter than 30 m, the average accuracy rate is higher than 90%, and the lowest accuracy rate is still above 80%.

Fig. 25 shows the accuracy rate with different WiFi duty cycles. Since the command recognition network is based on the MFCC spectrum and the NUCZT algorithm performs better in audio spectrogram recovery than the NUDFT and simple interpolation method, our method achieves the highest accuracy rate. Even when the duty cycle is only 20%, the average accuracy is still higher than 70%.

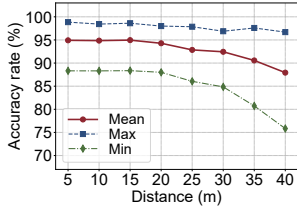


Fig. 24: The accuracy rate versus the Tag-to-Rx distance.

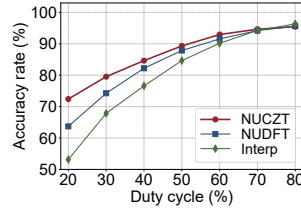


Fig. 25: The accuracy rate versus the WiFi duty cycle.

VIII. RELATED WORKS

With the development of IoT sensing systems, backscatter technology has gained attention as a low-power method for transmitting sensor data [5], [8], [9], [12], [14]. Based on how the sensor information is embedded, the backscatter can be classified into analog and digital types. We introduce related work of both categories and highlight the difference between Satori and these works.

A. Analog Backscatter

The analog backscatter directly embeds analog signals into reflected signals by amplitude, frequency, pulse, or phase modulation, avoiding digital processing.

The most related work is M.A.B. [24], which achieves multi-channel audio transmission through pulse position modulation. In our evaluation, M.A.B. achieves comparable SNR performance to Satori and even performs better at close range. However, the pulse position modulation is sensitive to self-interference, and any jitter in the carrier can be misinterpreted as a pulse. Therefore, M.A.B. requires a dedicated device to transmit a single-tone RF signal, which increases the deployment cost. In contrast, Satori embeds the audio into existing WiFi signals, avoiding this deployment cost. Moreover, to avoid interference from the in-band carrier, M.A.B. needs to shift the reflected signal to another blank band, resulting in higher power consumption and larger bandwidth overhead. In contrast, Satori avoids this issue through in-band transmission. Thus, the superiority of Satori lies in achieving comparable SNR performance, avoiding the deployment cost, additional spectrum occupation and high power consumption.

Similar to M.A.B., [12], [13], [24], [30], [31] also use pulse modulation to transmit analog signals, but they are all limited by the deployment cost of requiring dedicated devices to transmit clean carrier signals.

The amplitude modulation (AM) and frequency modulation (FM) are classic audio modulation methods that have been adopted by some analog backscatter works. For example, some works [32], [33] achieve AM by continuously adjusting the reflected signal's amplitude using a MOSFET. Specifically, [32] demonstrates a battery-free audio communication system through AM. However, the AM inevitably weakens the reflected signal's strength, and they do not handle the self-interference, leading to short communication ranges. There are also works that achieve FM by continuously adjusting the reflected signal's frequency based on the input analog

signal [34], [35]. However, these solutions require a high-power voltage-controlled oscillator (VCO) to modulate the frequency of the signal.

Researchers also explore directly embedding analog signals into the phase of the reflected signal [11], [36], [37]. In [11], the analog signal is embedded in the phase of ESS CSI in WiFi packets, enabling analog voltage signal transmission at the packet level. In contrast, Satori embeds the audio signal into the WiFi signal at the symbol level, enabling high-quality audio transmission.

B. Digital Backscatter

Digital backscatter first converts the sensor data into digital signals and then embeds them into the backscattered signal. Researchers have embedded digital signals into ambient carriers, such as WiFi [6], [14], [38]–[44], LTE [7], LoRa [8], [9], [45], [46]. Some digital backscatter works also transmit data over pilots or achieve in-band transmission. [47]–[49] use the pilot subcarriers as a single tone and shift them to another band, enabling the digital signal transmission. [18], [50]–[52] demonstrate in-band transmission. Specifically, [52] uses quasi-orthogonal OFDM codewords to eliminate self-interference. Compared with these works, Satori achieves in-band analog embedding and leverages the fixed pilot subcarriers to recover the analog modulated signals.

IX. CONCLUSION

This paper introduces Satori, an in-band analog backscatter system that enables ultra-low-power audio transmission. The Satori tag exploits analog modulation to embed the analog audio signal into every WiFi symbol directly, and transmits the audio inside the frequency band of the excitation, significantly reducing the power consumption on the tag. The Satori receiver eliminates the effect of CFO and frequency-selective fading, ensuring the quality of audio transmission. It also effectively recovers the audio spectrogram despite the bursty WiFi excitation causing audio sampling loss. From a broader perspective, Satori follows the paradigm of RF computing [10], [11], [15], [46], offering a direct manipulation from analog audio to the phase of RF signals. Our evaluation shows that Satori operates at a sampling rate of 41.67 kHz and achieves an SNR exceeding 18 dB for the transmitted audio in an indoor environment.

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